

$$T_\psi \ddot{\psi} + \dot{\psi} = K_\delta \delta + \frac{1}{N_{\dot{\psi}}} N_d$$

$$T_\psi = \frac{I_z}{N_{\dot{\psi}}} = \tau_1 + \tau_2 + \tau_3, \quad K_\delta = \frac{C_\delta}{N_{\dot{\psi}}}$$

$$\frac{\Psi(s)}{\Delta(s)} = \frac{K_\delta}{s(T_\psi s + 1)}, \quad \frac{\Psi(s)}{N_d(s)} = \frac{1/N_{\dot{\psi}}}{s(T_\psi s + 1)}$$